

Stable Disagreement with Markets: Research and Repeated Forecasts on New Contracts

Vorhersage experimental series

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Abstract

We extend a blinded model comparison to seven new unresolved contracts and three fresh calls per model-question-condition cell. The 126 calls cross GPT-6 Astra, Claude Fable 5.1, and Gemini 3.1 Pro with question-only and outside-research conditions. Market targets are committed before research and revealed after all responses are screened and sealed. Research reduces mean absolute distance from opening midpoints by 26–43%, but research distances of 18.11–22.92 percentage points exceed the constant-50% control’s 15.16 points. Average within-question research standard deviations are 1.16–3.28 points. Repeated calls expose relatively stable disagreement, particularly on disease incidence and global temperature; simple averaging offers little relief. The study also implements versioned output triage that distinguishes narrowly specified denials of market use from references requiring pre-reveal review.

1 Motivation and design

The preceding experiment suggested that current outside evidence can move language-model forecasts toward market assessments, while also producing shared mistakes and rankings sensitive to the question mix. It exposed a measurement problem: a lexical rule excluded responses that merely denied using prediction-market odds. This extension improves that rule, broadens topic coverage, and adds repetitions to distinguish variable responses from stable disagreement.

Prediction-market prices provide a timely intermediate benchmark for unresolved events. The comparison can be performed before outcomes occur, avoiding the use of realized outcomes as inputs and making repeated method development practical. Here, market distance measures agreement with contemporaneous assessments. It does not substitute for a later proper-scoring evaluation against outcomes, and a minimal book screen does not establish deep liquidity for every selected contract.

The protocol nominated eleven exact contracts from price-free definitions, with one threshold per event and no replacements after screening. Four failed the registered book requirements: TSA travel had insufficient top-of-book size; gasoline and Arctic ice lacked usable two-sided books; Rocket Lab launches exceeded the 10-point spread limit. Seven passed quote and outcome checks: mortgage rates, housing starts, measles, global temperature, Miami and Austin daily maximum temperatures, and a film-release delay. These span five declared broader dependence groups; housing and interest rates share a group, as do the two daily-weather contracts. This extends domains on the same collection date, not across multiple independent dates.

For each question, three models each receive three question-only calls and three research calls. Research calls receive the same frozen packet; they do not see the question-only answers or other

models’ forecasts. The method and provider configurations match the preceding comparison. An administrative replicate identifier distinguishes requests, caching is disabled, and the archive contains 126 distinct raw provider response identifiers. Repetition numbers do not imply common random draws across conditions.

2 Evidence, eligibility, and blinding

Evidence was collected after opening prices were sealed and before trial registration. It consists of official Freddie Mac weekly mortgage observations, the Census construction release, CDC case counts, NASA GISS temperature tables, NWS forecasts, and the dedicated official film site. The measles contract’s full terms clarified its U.S./CDC definition; this definition supplement was provided in *both* conditions, never treated as likelihood evidence.

The current CDC count was below the nominated threshold; NASA’s final four monthly values and annual value were missing; both weather target days had yet to begin; and the film site continued to advertise its scheduled release. Weather forecasts were required to be issued before the local target day, with a global forecast cutoff of 03:45 UTC on September 19. All calls and seals preceded that cutoff.

Each forecasting worker received only the event definition, instructions, and assigned evidence. No Kalshi research evidence, prices, browsing tools, prior-study results, or other forecasts were supplied. Verification compared all 126 returned model identities and actual system/user prompt receipts with registered inputs. All terminal attempts, screening records, and accepted forecasts were sealed before price revelation. Target commitments match the subsequently disclosed records. The coordinator’s separation from the evaluator was procedural, not independently enforced operating-system isolation.

The packets also preserve consequential access limitations. A current Miami forecast was available as an NWS search-index capture; two direct coordinate opens failed and another returned a September 13 page. The older page was not presented as the September 18 update. Weather settlement uses The Weather Company, so NWS point forecasts can differ in station/grid treatment and rounding. The climate packet supplied monthly observations but no ENSO or annual-temperature outlook; the disease packet supplied cumulative totals but no recent weekly incidence series. These omissions are visible to every model.

3 Measures and results

Let p_{imcr} be replicate r from model m , condition c , and question i ; let q_i be the sealed opening midpoint. The primary measure averages absolute loss within question before averaging equally over the common question set:

$$L_{mc} = \frac{100}{N} \sum_{i=1}^N \left\{ \frac{1}{R} \sum_{r=1}^R |p_{imcr} - q_i| \right\}, \quad N = 7, \ R = 3.$$

This estimates the descriptive average loss of an individual call on the selected cohort. A distinct secondary measure scores the three-call mean $\bar{p}_{imc}$: $100N^{-1} \sum_i |\bar{p}_{imc} - q_i|$. Their difference quantifies any improvement in absolute distance from averaging repetitions. We also report RMSE, distance outside the spread, per-cell sample standard deviations, and ranges. Repetitions do not increase the number of independent questions.

All 126 responses passed structural validation, identity checks, and screening, so every question is complete across all six model-condition cells and all three repetitions. No missing-value imputation, replacement calls, or manual response repairs were needed.

Table 1: Primary mean absolute distance from opening midpoints (percentage points). The matched cohort contains seven questions, not 126 independent events.

Model	Question only	Research	Reduction	Relative
GPT-6 Astra	28.54	18.11	10.43	36.5%
Claude Fable 5.1	25.54	18.82	6.71	26.3%
Gemini 3.1 Pro	40.01	22.92	17.10	42.7%

Research improves average agreement for all three models. However, the **constant-50% control scores 15.16 pp**, better than every model’s research forecasts on this cohort. The equal-weight average of the three research means scores 18.98 pp and does not outperform Astra. The constant control was present in the analysis implementation pinned at registration. The largest opening-to-follow-up midpoint movement is 1 pp; using follow-up quotes preserves the model ranking.

Table 2: Within-question variation and averaging. SD is averaged over questions; all values are in percentage points. It is not a confidence interval.

Model	Mean within-question SD		Research mean forecast MAE
	Question only	Research	
GPT-6 Astra	1.08	1.31	18.11
Claude Fable 5.1	5.44	1.16	18.82
Gemini 3.1 Pro	1.99	3.28	22.63

Averaging repetitions leaves Astra and Claude’s MAEs unchanged because their research draws lie on the same side of each target. Gemini’s mean-forecast MAE falls only from 22.92 to 22.63 pp. Low variability therefore should not be interpreted as well-calibrated confidence. The forecasts can be reproducible while remaining far from the external benchmark.

Two questions account for especially large shared gaps. For measles, the midpoint is 61%, while research means are approximately 24% (Astra), 7% (Claude), and 13% (Gemini). For global temperature, the midpoint is 74.6% and research means are approximately 32%, 32%, and 11%. In contrast, research substantially narrows the gap for the Austin temperature interval and film delay. Question-faceted slope graphs follow the appendix; their annotations score the *mean forecast*, while Table 1 reports the primary mean of individual-call losses.

4 Interpretation

This batch reinforces the value of supplying current information but weakens any claim that the present research method is already reliably competitive with a simple benchmark across topics. Absolute performance is worse than in the previous cohort, but the questions differ; that is not evidence that a model or the package deteriorated between runs. Rankings remain dependent on the selected questions and source packets.

The observed within-model variation is generally small relative to the large market gaps. A

useful next diagnostic is therefore targeted evidence variation: for example, recent weekly disease incidence and a relevant scientific annual-temperature outlook. The present experiment does not establish that these omissions caused the gaps; it identifies a testable explanation. Simply buying more draws of the same packet is unlikely to address stable disagreement of this magnitude.

No calibration function or learned ensemble weights were fitted. The seven questions and five broader groups do not support a general correction, and repeated calls cannot supply the missing event diversity. This cohort is now available for exploratory diagnosis, not untouched validation. Any correction or richer evidence policy should be frozen before testing on a later cohort, with identity calibration, equal weights, and the constant control retained. No significance claims or population confidence intervals are made here.

A Package changes and operational record

Two dependency-free Python helpers were added to Vorhersage. The versioned `market_screen.screen_market_mentions` returns lexical audit flags. Narrow, complete denials of using market information are labeled `denials_only`; qualified denials and other mentions require a recorded review before target revelation. Matching a word does not itself establish price exposure, and absence of a match does not certify isolation. The exception applies only to forecast prose; source-domain and source-content checks remain strict. Older studies retain their original registered acceptance rules.

The helper `market_evaluation.score_draws` computes expected single-call loss separately from loss of the mean forecast, along with sample SD, range, RMSE, and outside-spread distance. Missing or invalid values cannot silently enter as zero. A single draw has undefined sample SD, represented as `None`. Regression tests cover averaging across opposite sides of a target, missing repetitions, identity changes, denial-plus-price statements, and review decisions applied to the wrong response hash.

This batch had no market references in its 537 screened rationale/limitation fields, so no manual review was needed. It consequently does not estimate an acceptance-rate effect of the new rule. All 126 calls were accepted under the new policy, and 126 provider IDs establish distinct returned responses.

Experimental arms bind a method, model settings, and data policy. Immutable question versions and evidence packets are assigned to the six arms; the experiment expands each question/arm into three trials. Exact tasks and jobs are hashed before inference. Returned identities and prompts are checked, accepted forecasts are issued as immutable artifacts, and all attempts remain in cost accounting. A separate seal binds screening and any review decisions to the raw results and forecast seal. The workbench reveals its committed opening targets only after these checks complete. Independent verification recalculates the published scores and checks chronology and target commitments.

Astra uses high reasoning effort; Claude uses adaptive thinking with high output effort; Gemini uses a 2,048-token thinking budget and temperature 0.5. Each has an 8,192-token configured output limit. These are provider-specific settings, not equal compute budgets. Exact serialized settings and raw provider responses are archived. Job-level and per-response cost sums differ slightly by rounding. The full package suite passes 346 tests; two upstream deprecation warnings are unrelated to this change. The database integrity check passes for 547 artifacts.

Table 3: Reported inference cost, including all attempted work. Research/coordinator cost is unknown and excluded.

Model	Calls	USD
GPT-6 Astra	42	1.359340
Claude Fable 5.1	42	3.021870
Gemini 3.1 Pro	42	0.637074
Total	126	5.018284

B Source and artifact index

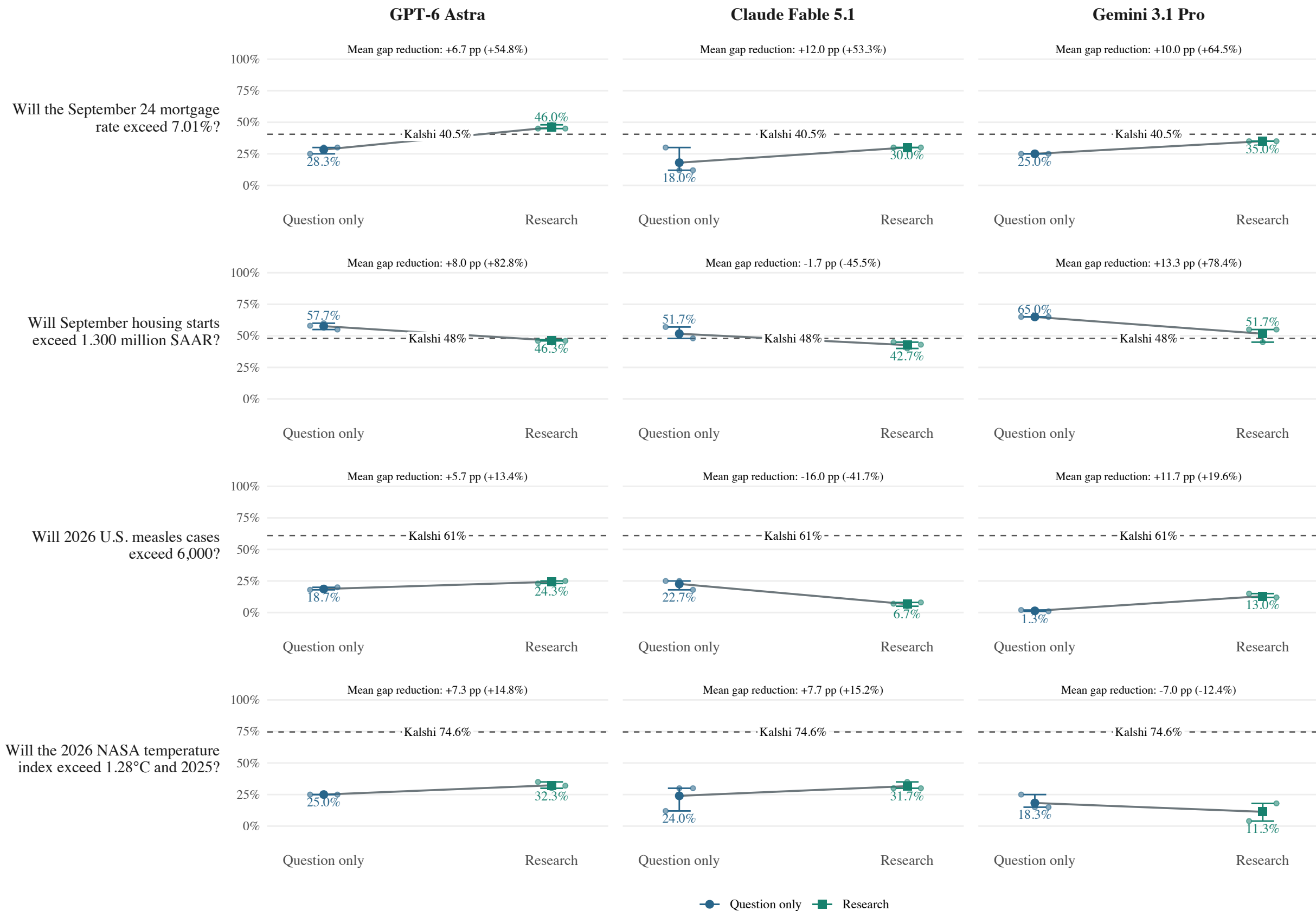
The source receipts, failed opens, frozen packets, definitions, protocol, six arms, 126 tasks, provider jobs/results, screening records, seals, and revealed targets are preserved under `examples/kalshi_repeated_20260918/`. The accompanying README gives the reproduction commands. `run/comparison.json` contains the scores; `verify_results.py` checks them without new inference or private-vault access. `plot_performance.R` generates the following vector plates with ggplot2. Their bars show the observed three-draw range, not uncertainty intervals for a latent true probability.

Primary source routes are the [Freddie Mac PMMS archive](#), [Census construction release](#), [CDC measles counts](#), [NASA GISS temperature analysis](#), [NWS point forecasts](#), and the [official Dune film site](#). Dynamic pages may change; the saved receipts and retrieval timestamps define this study’s evidence.

Research and repeated forecasts: part 1 of 2

Three fresh calls per condition; lines join means; small points and bars show draws and their range

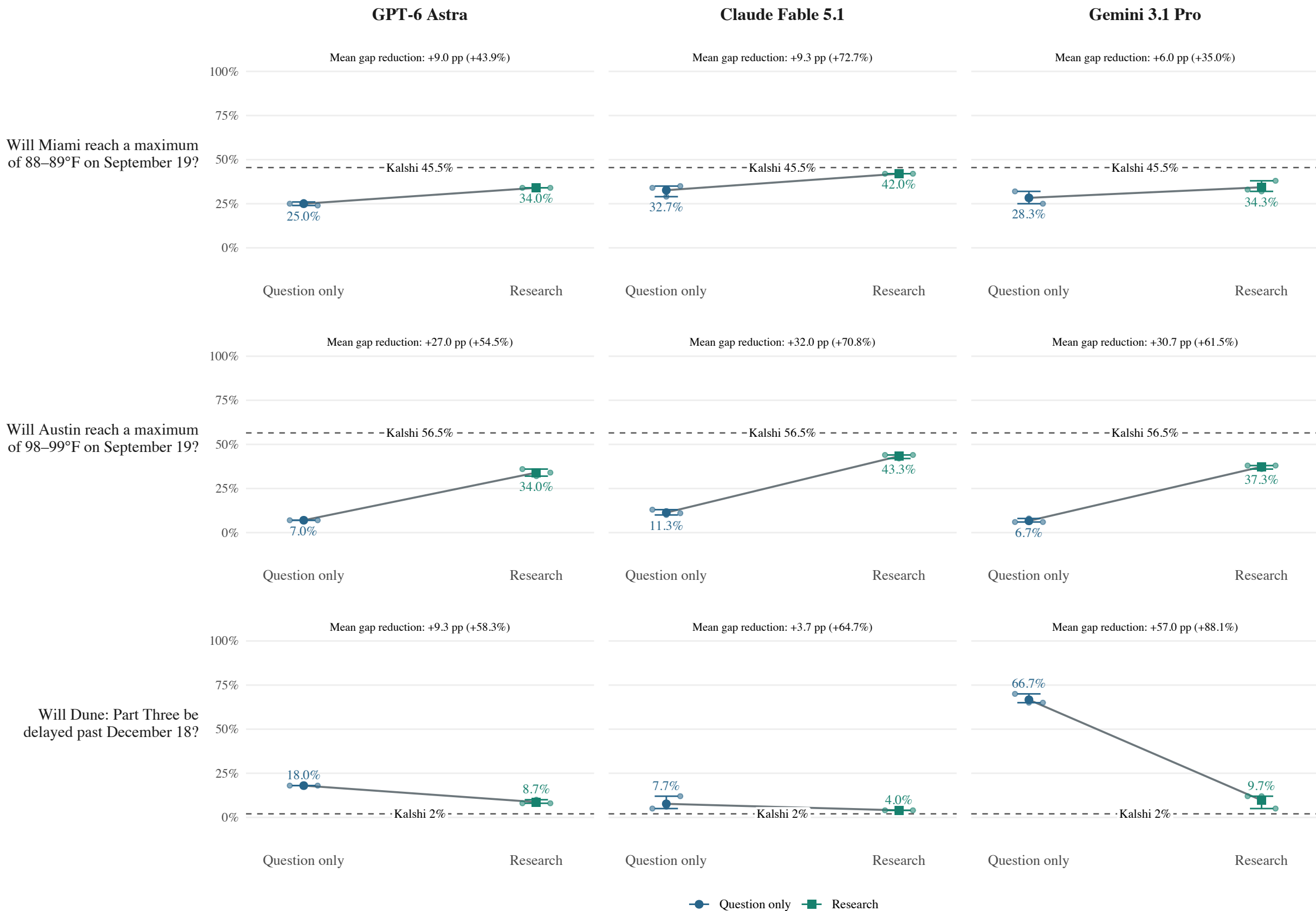
Forecast probability



Dashed line: opening midpoint, sealed before research. Bars are observed ranges, not confidence intervals.
Annotations score the mean forecast; the primary table averages losses across calls within each question.
Question-only and research calls are separate; replicate numbers do not imply matched random draws.

Research and repeated forecasts: part 2 of 2

Three fresh calls per condition; lines join means; small points and bars show draws and their range



Dashed line: opening midpoint, sealed before research. Bars are observed ranges, not confidence intervals.
Annotations score the mean forecast; the primary table averages losses across calls within each question.
Question-only and research calls are separate; replicate numbers do not imply matched random draws.